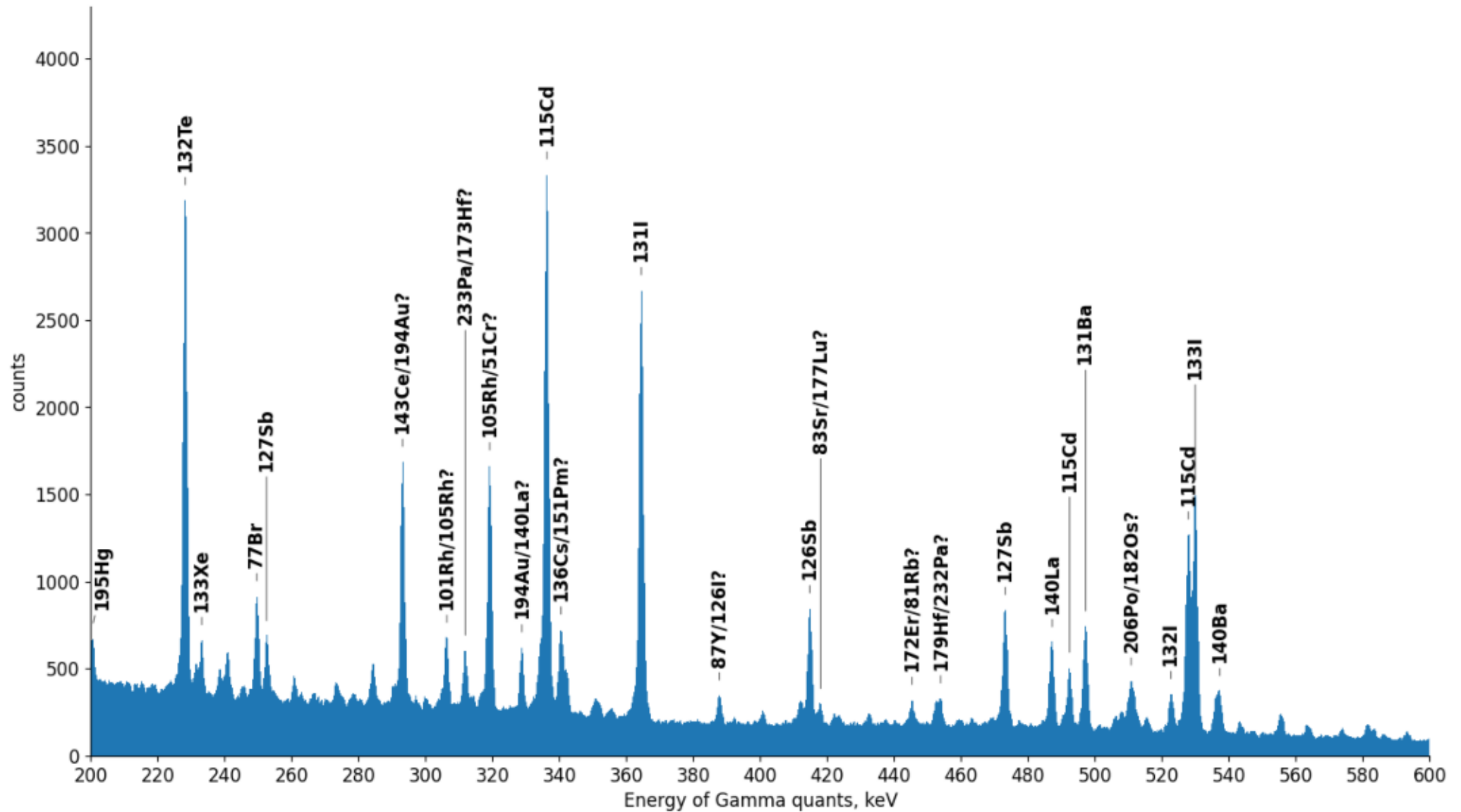


Gamma Spectrum Analyzer — sample outputs

All figures produced by `analyze_spectrum.py` from the bundled `test_spectrum.Spe`: a 1083 s acquisition on an Ortec MCB, taken 5.04 days after a 119-minute proton irradiation.

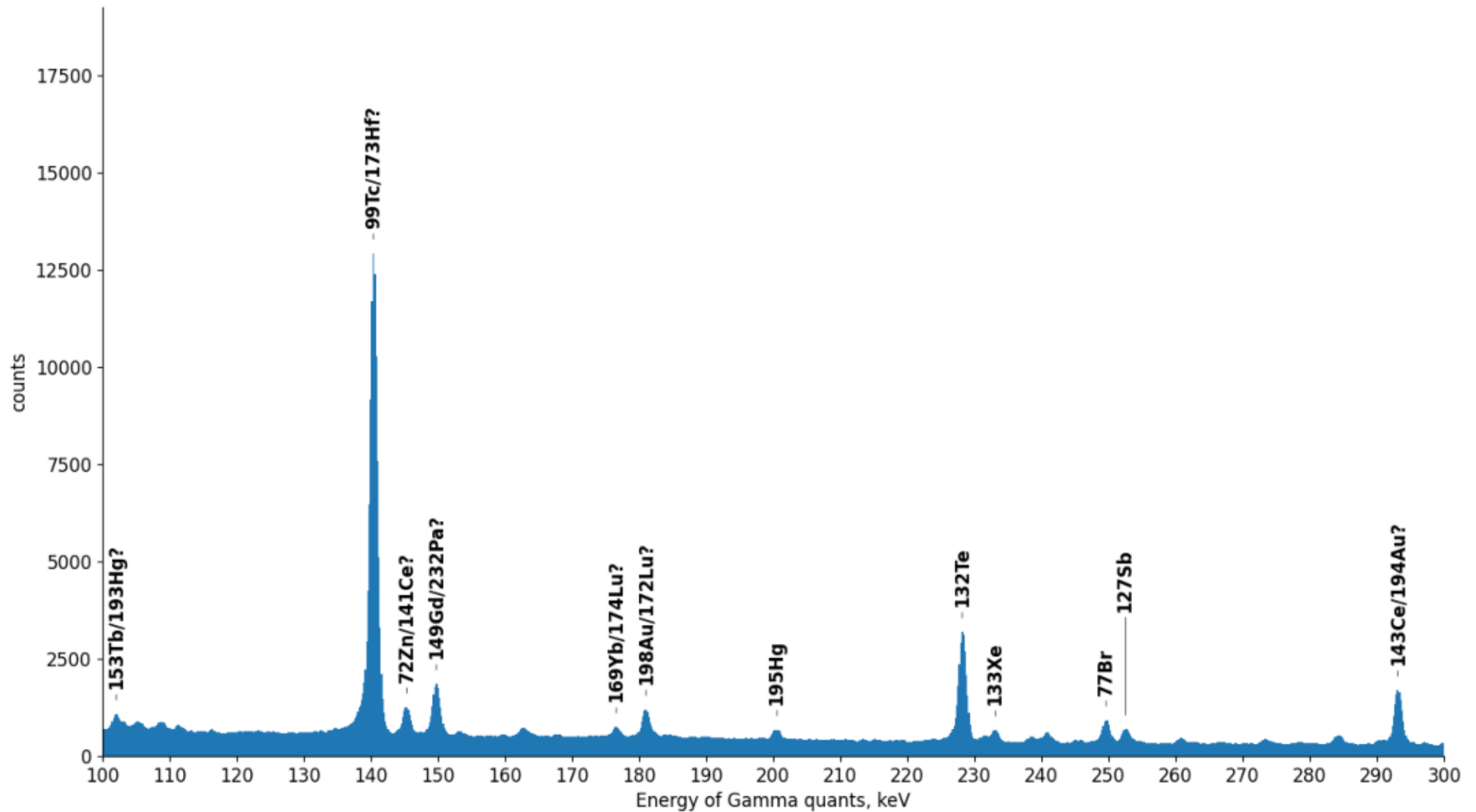
The energy axis comes from the `MCA_CAL` calibration stored in the spectrum file (0.191 keV/channel, 16384 channels).

Summary spectrum, 200-600 keV



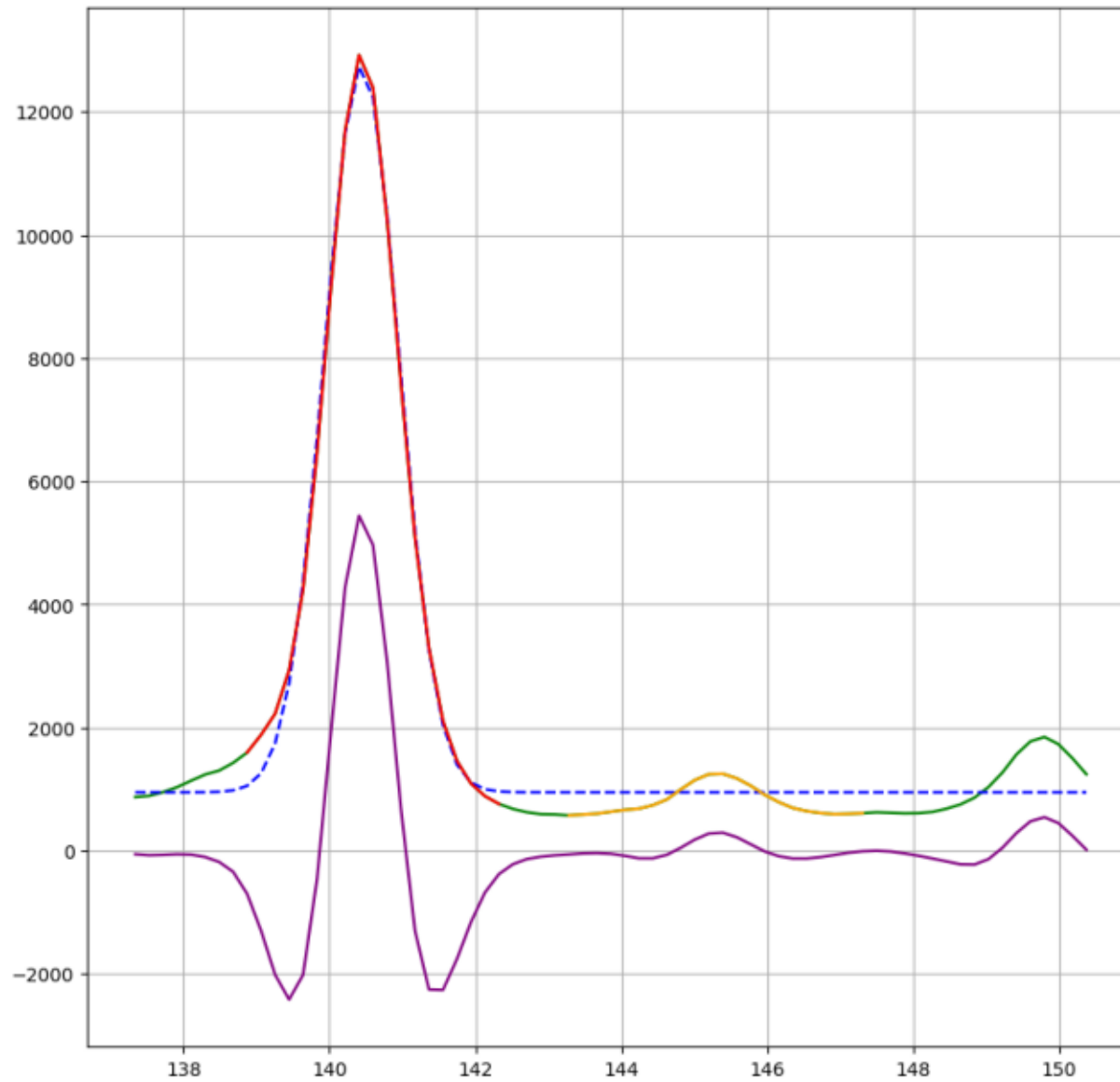
Peaks found by the convolution filter, each labelled with the most plausible gamma-library isotope. Two candidates and a trailing "?" mark an ambiguous match. Strong lines: 132Te (228 keV), 115Cd (336 keV) and 131I (364 keV) — ordinary fission/activation products.

Summary spectrum, 100-300 keV



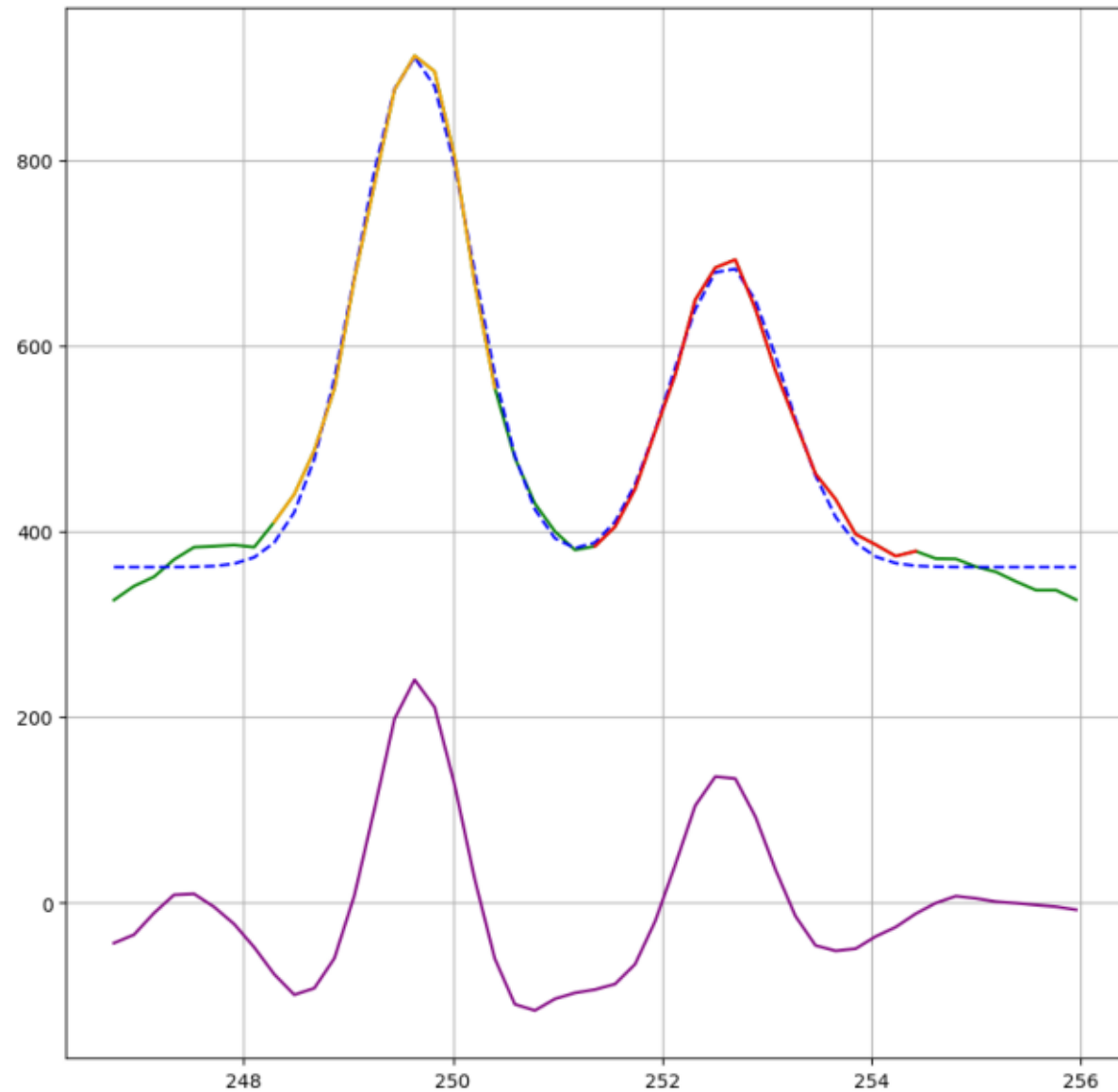
The same spectrum over a narrower window, dominated by the 140.5 keV line of 99Tc (the Tc-99m decay gamma), the strongest peak in the file.

Single-peak Gaussian fit — 140.5 keV (99Tc)



Red: smoothed counts in the fitting window. Purple: convolution-filtered spectrum used to set the window boundaries. Dashed blue: fitted Gaussian. The net peak area gives 72.8 counts/s and an activity of 7.05e6 decay/s for the configured sample parameters.

Overlapping-peak fit — 249.6 / 252.5 keV (77Br / 127Sb)



Two peaks closer than the detector resolution are fitted with a double Gaussian. Orange and red: the two single-peak windows. Green: the joint window. Dashed blue: the eight-parameter joint fit. Purple: filtered spectrum. The lower curve shows the background-subtracted residual used for the area integration.